



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (25/05/2026 – 30/05/2026)

WEEK 2

PROJECT NAME

CHANGE DETECTION

PROJECT DESCRIPTION :

This project aims to detect changes between satellite images captured at different times using Deep Learning and Computer Vision techniques.

GROUP MEMBER :

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GROUP ID :

2026G266

GROUP GUIDE :

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COLLEGE NAME :

GATI SHAKTI VISHWAVIDYALAYA

25/05/2026	Explored the OpenCD framework for satellite image change detection. Studied the architecture and implementation of transformer-based models available in OpenCD.
26/05/2026	Installed and configured the OpenCD environment. Understood dataset structure and preprocessing requirements for model training.
27/05/2026	Worked with Google Earth Engine to acquire multi-temporal satellite imagery. Exported satellite images for further processing in QGIS.
28/05/2026	Holiday (Bakri Eid)
29/05/2026	- Performed image preprocessing and alignment using QGIS. - Generated sample datasets for change detection experiments. - Creation of dataset with GEE and QGIS.
30/05/2026	Studied GeoTIFF format and its importance in geospatial applications. Prepared image pairs suitable for change detection analysis.
31/05/2026	Holiday (Sunday)

Next Week Plan	Fine-tune a selected change detection model using the prepared dataset. Perform inference on satellite image pairs and evaluate results. Explore accuracy assessment metrics and visualization of detected changes.
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REFERENCE:

- <https://github.com/likyoo/open-cd/tree/main/configs/fcsn>
- https://developers.google.com/earthengine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED

SCREEN SHOTS:

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```
[ ] Map = geemap.Map()

rgb_vis = {
  'bands': ['B4', 'B3', 'B2'],
  'min': 0,
  'max': 3000
}

Map.addLayer(before, rgb_vis, 'Before')
Map.addLayer(after, rgb_vis, 'After')

Map.centerObject(region, 10)

Map
```



